

Coursework cover sheet**Section A - To be completed by the student**

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Semester: APRIL 2026	
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Module Code and Title: 5023CMD Introduction to Artificial Intelligence	
Assignment No. / Title: Coursework 1	50% of Module Mark
Hand out date: 17 th April 2026	Due date: 17 th May 2026
Penalties: No late work will be accepted. If you are unable to submit coursework on time due to extenuating circumstances, you may be eligible for an extension. Please consult the lecturer.	
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Section B - To be completed by the module leader

Intended learning outcomes assessed by this work (5023CMD):

1. Adhere to the social and legal implications of artificial intelligence. (A4, PLO11).
2. Identify the various approaches to knowledge representation in artificial intelligence. (C4, PLO1)
3. Assess problem-solving strategies under uncertain conditions, maintaining control and accuracy in execution. (P5, PLO3)
4. Illustrate an understanding of knowledge-based systems, and their respective architectures. (C4, PLO1)
5. Demonstrate artificial intelligence approaches in practical tasks to determine the most effective solution for an intelligent application. (P5, PLO3)
6. Verify human factors and the impact on Artificial Intelligence. (A5, PLO10)

Marking scheme	Max	Mark
1. Assignment 1	100	
Total	100	
Lecturer's Feedback		

Internal Moderator's Feedback

STUDENT DECLARATION OF GENERATIVE AI USE IN ACADEMIC WORK - INDIVIDUAL

This form must be submitted together with your assignment to declare the extent of AI usage, if any. Inaccurate or false declarations may result in a breach of academic integrity.

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Course Code: 5023CMD	Course Title: Introduction to Artificial Intelligence	Lecturer / Instructor's Name: Ms Vasuky Mohanan	
Assessment Title: Coursework 1		Date of Submission: 17/5/26	

A. AI Usage Declaration *(Please tick one):*

Category 1: GenAI Tools Used in Content Creation And/or Proofreading



I confirm that I have used GenAI tools (e.g., ChatGPT, Grammarly, Gemini) to support the completion of this assignment. I have described how GenAI tools were used in this assignment. In-text citation was provided whenever GenAI generated content was used.

☐

Category 2: GenAI Tools Used for Proofreading Only

I confirm that I have used GenAI tools or services only for proofreading purposes. No content in this assignment was generated by GenAI.

☐

Category 3: No GenAI Tools Used

I confirm that I have not used any GenAI tools or services, including for proofreading, in this assignment.

B. Details of AI Usage *(Only for Category 1 or 2):*

Section of Assessment	Page Number	GenAI Tool Used	Purpose of Use	How AI Output Was Modified
Task 2	Page 19	ChatGPT & Gemini	Concept Generation	Rewritten the draft proposed by Gemini & ChatGPT in my own words and added citations and references.

Acknowledgement

I understand that any undeclared or misrepresented use of AI may be considered a breach of INTI's Academic Integrity Policy and may lead to disciplinary action. I acknowledge that I am responsible for the work submitted and may be required to demonstrate the originality of my work or explain how GenAI was used, if requested by the lecturer/instructor or academic authority.

Student Signature: _____



Date: 17/5/26

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List of Abbreviations

i. ADMP	Automated Decision Making and Profiling
ii. AI	Artificial Intelligence
iii. AIGE	National Guidelines on AI Governance & ETHICS
iv. BIPA	Biometric Information Privacy Act
v. CAMA	Context-Aware Moderation and Assessment
vi. DPIAs	Data Protection Impact Assessment
vii. ETS	Educational Testing Service
viii. EU	European Union
ix. GCSE	General Certificate of Secondary Education
x. GenAI	Generative AI
xi. LLMs	Large Language Models
xii. MIDA	Malaysian Investment Development Authority
xiii. MQA	Malaysian Qualification Agency
xiv. NAIO	The National AI Office
xv. NLP	Natural Language Processing
xvi. OECD	Organisation for Economic Co-operation and Development
xvii. PDPA	Personal Data Protection Act
xviii. PDP	Department of Personal Data
xix. QWK	Quadratic Weighted Kappa
xx. TML	Traditional Machine Learning
xxi. UK	United Kingdom

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Student Assignment Brief

Task 1 (1903 Words): AI in education (admissions, grading, student monitoring)

Question

Critically analyze the following:

1. Social implications

- Who benefits and who may be disadvantaged?
- Issues of bias, inclusion, or inequality

2. Legal implications

- Accountability, liability, explainability
- Compliance with data protection or AI regulations

3. Ethical implications

- Transparency, fairness, human oversight
- Automation vs human judgment

Your analysis **must include both positive and negative impacts**, supported by real examples or reported cases.

Finally, **briefly discuss the relevance of this AI application to Malaysia**, and suggest **one realistic safeguard or policy measure** that could reduce harm.

Content

1.0 Introduction



Figure 1: Artificial Intelligence (AI) In Education: Transforming the Future of Learning (Bhatt, 2023)

As per Doss et al., (2025), the usage of AI is skyrocketing in the education sector, particularly in high-stakes environments such as student admissions, assessment, and monitoring. For instance, AI admission tool, AI-based writing evaluation tools, and remote proctoring systems like ProctorU aim to expedite the analysis of large data with increased consistency (Proctorio, 2026; Sun, 2026). Furthermore, the emergence of generative AI platforms like ClaudeAI and Gemini further expands the use of AI in assessment and learning.

Despite maximizing scalability and reliability, UCW (2025) highlights that suggestions provided by AI can pose problems by influencing the students' learning skills, and futures, raising issues of fairness, accountability, transparency, and human judgement. Therefore, this paper discusses the social, legal, and ethical issues of AI in education, including its pros and cons through real-world cases, and lastly analyzing the risks in Malaysia and how safeguards can be applied.

1.1 Social Implications

1. Speed vs Comprehensiveness

QWK	Interpretation	Human Grader Level
0.81 - 1.00	Almost perfect agreement	Expert Grader
0.61 - 0.80	Substantial agreement	Average Grader
0.41 - 0.60	Moderate agreement	Poor Grader
0.21 - 0.40	Fair agreement	
0.01 - 0.20	Slight agreement	
< 0	Less than chance agreement (Completely wrong!)	

Figure 2: QWK scores: Automated grading system efficacy (Kenehan, 2025)

Rapid advancements of GenAI and LLMs in recent years have streamlined analytical and evaluative tasks to (INFOLKS, 2026). According to Islam (2025), grading tools powered by LLMs can have an agreement of 85% with human evaluators via statistical metrics like **QWK**– which measures how closely two graders agree on score, indicating a high degree of consistency in closed assessment tasks (Kenehan, 2025). An actual example of this is the **e-rater scoring engine** developed by ETS that automatically scores essays and gives feedback on writing, demonstrating how AI can facilitate faster, more scalable assessment.

Nevertheless, this comes at the expense of evaluation quality. While the e-rater and similar programs are useful for scoring essay-oriented writing, ETS have pointed out that these systems are still in development to better evaluate more complex aspects of writing like argument structure and creative use of language (ETS, 2026). This demonstrates that automated grading still lacks the capacity to assess more nuanced elements of writing like critical thinking, creativity, and context (Han and Zheng, 2026).

2. Algorithmic Bias vs Consistency



Figure 3: Biased Algorithms Are Easier to Fix Than Biased People (Mullainathan, 2019)

According to Bernik, Radošević and Čep (2025), most recent AI models like Claude and Gemini exhibit refined grading consistency since they apply a standardized assessment approach. Thus, research has indicated that these tools reduce rater variability compared to humans, leading to more consistent marks (Han and Zheng, 2026). One reported case that demonstrated the appeal and danger of algorithmic standardisation is the **Ofqual grading controversy** in the UK, where student grades were algorithmically calculated after A-Level and GCSE examinations were called off during the COVID-19 pandemic (Kelly, 2021).

Such an approach, however, comes with risks, with this one being bias. With the Ofqual case, many results were 40% lower than those predicted by teachers, and complaints were directed at the model which worked against some lower socio-economic groups of students and in favour of certain private school groups (Cordella and Gualdi, 2026). Subsequent criticism triggered the removal of algorithmic grades and teacher-assessed grades were reintroduced, indicating that while AI can be reliable and standardized, it may also generate one-sided results if historical or institutional data used by the model is reflective of social inequalities (Yulianti, 2025).

3. Opportunity Expansion vs Admissions Inequality



Figure 4: Inequalities in College Admissions (Pogozelski, 2023)

According to EduTech (2026), universities can streamline processing large volumes of applications by leveraging AI-powered admission tools. By using algorithms such as NLP and TML, these automated screening platforms can help admissions teams by screening documents, ranking applicants, and identifying those who fulfil the entrance requirements (Jiang et al., 2025). This helps save time in administration and speeds up applicant response times, particularly in the period of high admission numbers.

However, Jiang et al. (2025) also noted that AI admissions systems can be prone to inequality if they heavily depend on past data or limited metrics to predict student success. A case study on models for undergraduate admissions revealed that these systems can be detrimental to the admissions process if bias is not appropriately addressed (Vasquez et al., 2025). This is mainly because the admissions data could be a result of existing inequalities based on factors like school background, first-generation status or access to academic preparation (Jiang et al., 2025).

4. Academic Integrity vs Privacy

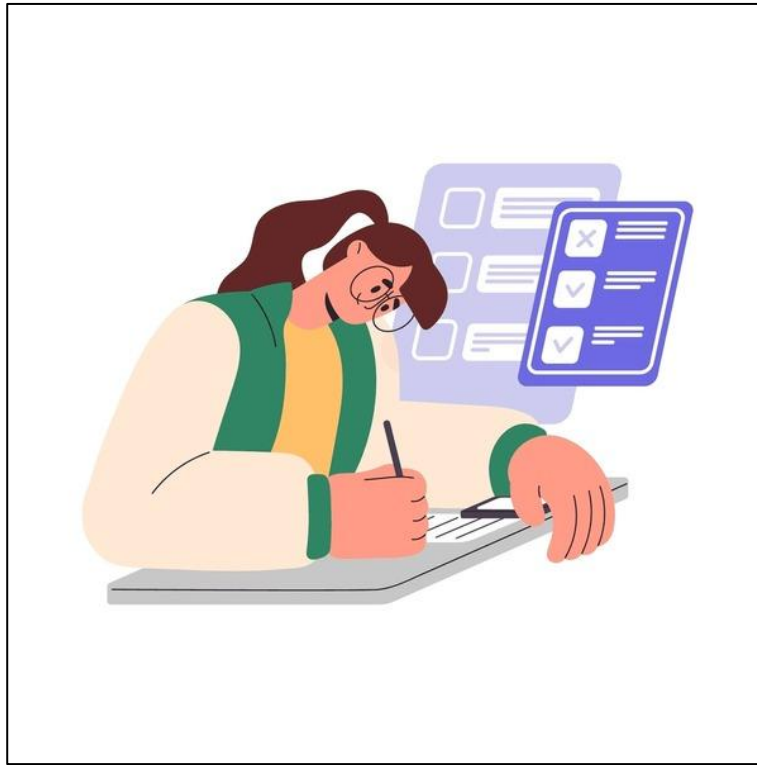


Figure 5: Academic Integrity (Shutterstock, 2022)

Universities are increasingly using tools such as **Turnitin AI detection** and **Proctorio** to prevent academic dishonesty through AI-assisted cheating (Young, 2025). One such case is **ProctorU's biometric privacy lawsuit**, in which students claimed the online proctoring software improperly collected and stored their biometric data, violating the Illinois' BIPA (Shaak, 2021).

Nonetheless, extensive environmental monitoring can result in social exclusion. This is due to the system's assumption of a quiet and sterile testing environment and therefore punishes students from lower socio-economic backgrounds that lack environmental privilege (Kwapisz et al., 2025). Those living in communal and compact spaces are likely to be marked as background noise and/or the presence of family members (Wilson and Shankar, 2021). This leaves a huge equity gap as a student's success in high-stakes assessments is based on their living conditions, rather than academic merit (Yoder-Himes et al., 2022).

1.2 Legal Implications

1. Liability for Harm vs Absence of Clear Legal Responsibility



Figure 6: Top 10 Surprises of Accountability (Rhoads, 2024)

Grading and evaluation tools like Gradescope can simplify decision-making and decrease human error, which may potentially enhance institutional efficiency (Hansel et al., 2024). When results are satisfactory, it can lead to quicker and more efficient processes for institutions. Nonetheless, when things go wrong, such as inaccurate grading or accusations of cheating, legal responsibility becomes unclear.

Because unlike conventional human analysis, the results of AI assistance boil down to software engineers, educational institutions, and lecturers (Mita, 2023). For instance, a Yale University student who filed a lawsuit after he was reportedly suspended for allegedly committing a “misconduct” that involved AI (Zahn, 2025). The case illustrates the legal complexities that can arise when academic sanctions are based on AI-generated evidence or detection (Crowell, 2025).

2. Right to Appeal vs Limited Contestability of AI Decisions



Figure 7: Right to Contest (Mit.edu, 2026)

The standardization of decision-making processes by AI platforms can help facilitate fairness and consistency within large student groups (Chaudhary, 2025). Online services like Turnitin's Feedback Studio offer universities with high-quality similarity reports that can be used to support their decisions (Turnitin, 2025).

But legally, individuals are entitled to appeal against decisions that jeopardize their grades. Despite this entitlement, Kosinski (2024) states that certain results created by AI can be hard to challenge as the logic it applies are often ambiguous and hard to comprehend. An Adelphi University student successfully challenged his AI plagiarism accusation in court and had the plagiarism removed from his record, demonstrating the need for students to challenge academic decisions made with the use of AI (Kaminski, 2021).

3. Adherence to Data Protection Laws vs Risk of Non-Adherence



Figure 8: A comparison of data protection laws (Asia Business Law Journal, 2022)

AI systems are dependent on the large amount of data they can gather to work effectively, especially in monitoring tools like ProctorU and ExamSoft and AI-powered admissions platforms, such as Liaison Othot. When properly developed, such systems can work within a legal infrastructure because the data is gathered and computed lawfully.

However, serious legal repercussions may arise for businesses that fail to comply with these requirements. In admissions process, this risk is great because applicants submit personal information, academic records, financial data, and supporting materials (leadersinlaw, 2026). A case in point being a College Board deal in New York, where it was forced to pay \$750,000 after it was accused of illegally licensing personal student information (Stempel, 2024). Although unrelated to self-scoring AI, it illustrates the potential for legally harmful results when AI in education-related data around admissions is misused.

4. Regulatory Development vs Legislative Uncertainty



Figure 9: The European Union Artificial Intelligence Act: Bridging ethics, innovation and governance (Parva Consulting, 2025).

Given the growing prevalence of AI in education, both local and international bodies have come up with more streamlined regulations to foster responsible application. According to the European Commission (2026), risk-based strategies like the EU AI Act can classify AI systems according to the level of harm that they may cause, with more demanding conditions applied to high-risk applications like those in education. Furthermore, international models like the OECD Principles on AI focus on transparency, and human supervision, promoting ethical and legally responsible AI practices within institutions (OECD, 2026).

In spite of these developments, there is still a lot of legal uncertainty. Most present-day laws are not created to deal with AI-related issues like automated decisions, profiling, or the application of probabilistic models in high-stakes scenarios (Unesco, 2026). This means that schools can be working in grey areas in terms of regulations, as the requirements are not well outlined. Therefore, the disparity between technology and legislative development presents compliance concerns and emphasizes the necessity of more thorough, AI-specific laws geared to the education sector.

1.3 Ethical Implications

1. Transparency vs Opacity

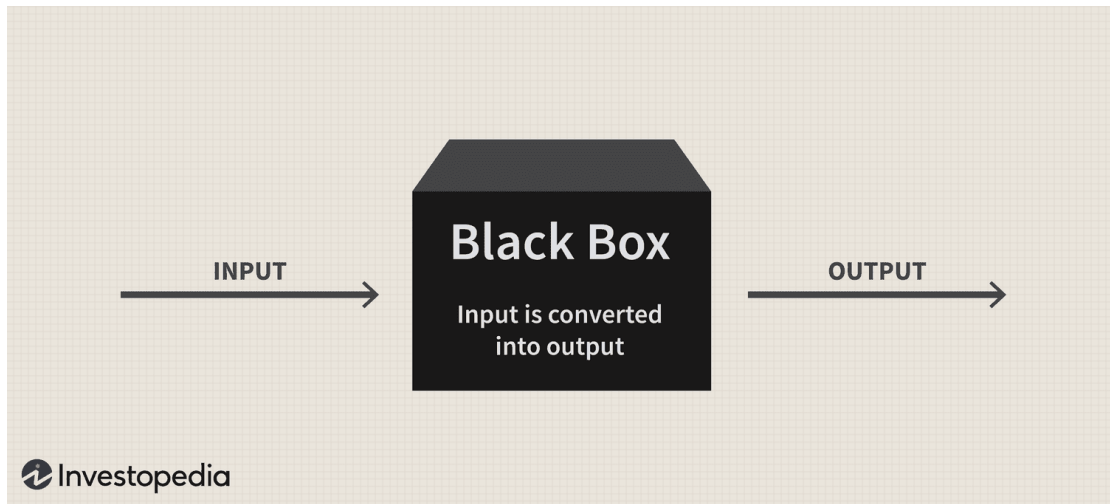


Figure 10: What Is a Black Box Model? Definition, Uses, and Examples (Kenton, 2025)

According to Rubman (2024), newly developed LLMs and AI platforms have enhanced the efficiency of grading and evaluation processes. Google Gemini can deliver real-time grades and detailed feedback, which can prove to be useful when marking the exams of hundreds of students (Gemini, 2025).

However, these systems often function as black box models, in the sense that their decision-making methodologies are usually complex and vague (Kosinski, 2024). This concern was raised within Vanderbilt University's decision to turn off Turnitin's AI detector, as the university lacked a proper understanding of the tool and how students would be impacted by false positives (Diplo, 2023). As per Karthiga (2025), this lack of openness causes students and educators to ineffectively question or scrutinize decisions, putting forward an additional ethical concern of whether AI should make high-stakes educational decisions that cannot be clearly outlined.

2. Autonomy vs Overreliance



Figure 11: Start-ups: Over reliance on technology (Jhunjhunwala, 2026)

By incorporating Artificial intelligence in education, students are provided with tailored recommendations, and learning assistance. Khanmigo and ChatGPT are some of the tools that enable students to refine their work independently, potentially improving the learning process (Yamkovenko, 2024).

However, this raises an ethical concern of student autonomy and excessive dependence on AI. As a result, 59% of students fear they may lose the ability to think critically and solve problems on their own since they will be more inclined to use AI-generated suggestions (Turnitin, 2026). Such limitations raise questions about whether AI promotes education or disrupts the development of fundamental intellectual abilities.

3. Institutional Power vs Student Agency.



Figure 12: Authoritarian regime flat vector illustration (Vecteezy, 2026).

AI-driven admissions can support universities in making better decisions, as it can help them identify the likelihood of prospective students enrolling, remaining in school, and succeeding (Watermark Insights, 2024). Predictive analytics tools like **Liaison Othot** offer live insights and suggestions for recruitment and enrolment planning, allowing institutions to effectively handle huge applicant pools (Liaison, 2026).

Nonetheless, Bird et al. (2021) insists that this poses ethical issues regarding student autonomy. Students may be completely unaware that their applications are ranked, filtered, or prioritized by an AI-based admission system. Predictive analytics in admissions have been discussed at length on how colleges can rate students based on their chances of applying, getting accepted, and enrolling. But when applicants cannot comprehend and contest these scores, it causes a power shift in favour of institutions and a reduction in control for the applicants.

4. Surveillance vs Privacy



Figure 13: How Data Privacy Concerns Impact Firm Performance (Wharton, 2023)

AI-based Monitoring and evaluation enhance the capacities of the institutions to sustain academic integrity and deliver unbiased grading (Tan et al., 2025). ProctorU and ExamSoft are tools that apply facial recognition, behavioural analysis and environment monitoring to identify suspicious behaviour in the course of examinations (ExamSoft, 2025; ProctorU, 2025).

However, such proctoring tools requires collection of sensitive biometric and environmental information. A reported example is the French TestWe case, where an administrative court stopped a university's use of an e-proctoring platform when students questioned its facial detection and algorithmic surveillance capabilities (Gullo, 2023). Other than being algorithmically biased, Miller and Manderfeld (2021) emphasised that these algorithms can be culturally and religiously biased too; for example, if a student wears a niqab, they might be marked as an outlier due to deviations from the AI's training data.

1.4 Relevance to Malaysia and Proposed Safeguard

According to NAIIO (2025) and MIDA (2025), Malaysia's adoption of AI in education is in line with the National AI Roadmap (2021-2025) and the National AI Technology Plan (2026-2030). While the PDPA 2010 does not directly address ADMP at this moment, the ADMP Guideline 2026 seeks to fill this gap by incorporating principles for processing personal data and suggesting safeguards such as notices, DPIAs and human review (PDPD, 2026). To further strengthen in this direction, Malaysia can incorporate a human-in-the-loop safeguard into its proposed AI Governance Bill. This policy ensures that AI is only used as "recommender systems", and not as final arbiters for high-stakes academic decisions like final grading or expulsion due to AI proctoring (Selvam and González Vallejo, 2025).

Furthermore, it reflects the **right to interact with a human** and the **right to an explanation** principles outlined in the AIGE Guidelines (Securiti, 2024). Mandating the MQA to audit institutions for human oversight will help Malaysia move towards a balance between technological efficiency and ethical accountability so that black-box algorithms never outweigh a student's right to procedural fairness.

Task 2 (1899 Words): Context-Aware Moderation and Assessment (CAMA) system

Question

Based on the **same domain chosen in Task 1**, propose a **conceptual AI system** that improves decision-making **while reducing ethical and social risks**. You are **not required to implement or code** the system

Explain and justify:

1. Knowledge Representation
 - What type(s) of knowledge are required?
(e.g. rules, ontologies, probabilistic knowledge, cases)
 - Why are these suitable for the problem?
2. Reasoning or Planning Approach
 - How does the system make decisions under uncertainty?
 - What role does human oversight play?
3. Search or Inference Techniques
 - What AI search or inference techniques are appropriate?
 - Why are they suitable for this context?

Content

2.0 Introduction

The **Context-Aware Moderation and Assessment (CAMA)** system is a conceptual AI decision-support system for higher education assessment. It can assist in essay grading, academic integrity review and assessment in moderation in remote assessment moderation. The system addresses the core concerns of task 1, which were black-box grading, algorithmic bias, inequitable proctoring flags, restricted opportunities to appeal, concerns about privacy, and overreliance on automated decision making. CAMA is not a substitute for lecturers, it instead offers suggestions, justifications, uncertainty scores, and contextual evidence that supports said justifications. Final decisions are still human controlled, making the system appropriate for high-stakes learning as students’ grades, disciplinary action and appeals can significantly impact their education.

2.1 Knowledge Representation

2.1.1 Types of Knowledge Required

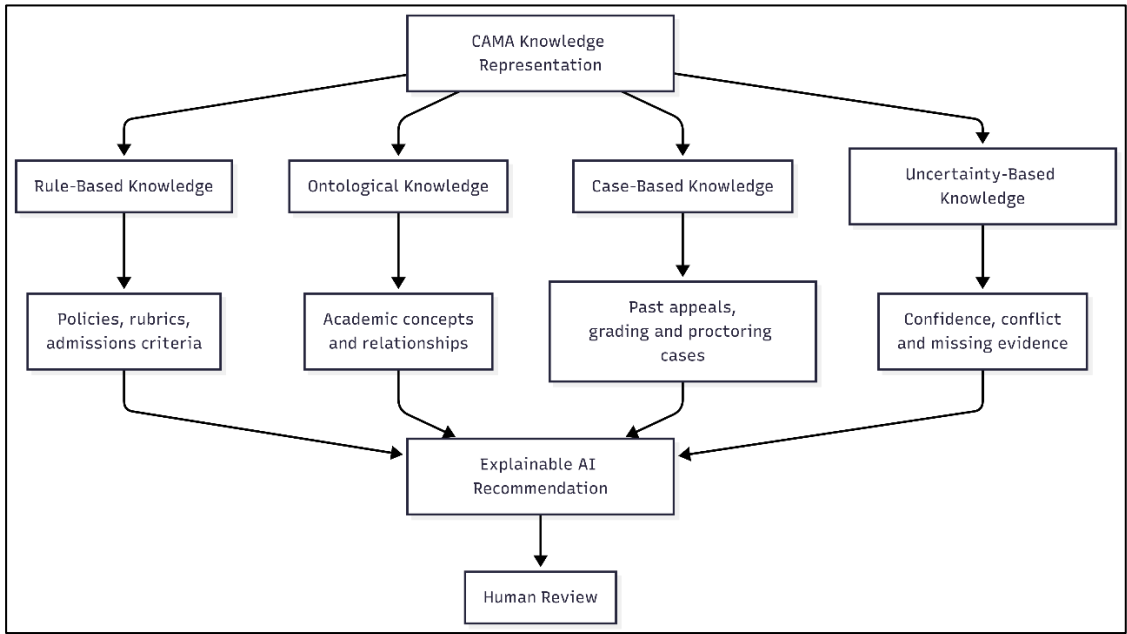


Figure 14: Four-Tier Knowledge Representation in CAMA

As shown in *Figure 12*, A hybrid knowledge representation model is required for CAMA, as one type of knowledge is insufficient for sustaining fair academic judgement. It has to weigh in formal academic requirements, institutional rules, student context, previous cases and the uncertainty of AI outputs. Thus, CAMA employs a four-tier knowledge model, namely the **Rule-based, ontological, case-based, and uncertainty-based knowledge**.

Firstly, **rule-based knowledge** includes the policies for assessment, grading rubrics, admission criteria,

procedures for academic integrity, privacy policies, appeal policies, and human review policies (Shah, 2023). An example of this may be that an AI-generated cheating flag does not automatically mean disciplinary action. A second rule may be applied where low-confidence grading recommendations should be presented to a lecturer. These rules matter as they prevent the system from being the ultimate decision-maker and maintains CAMA's role as a mere support system.

Ontological knowledge, often illustrated by a knowledge graph, describes important academic terms and their relationships (Popov, 2026). For instance, when an assessment is turned in, that assessment is evaluated by a rubric, and the criteria within that rubric are tied to the learning outcomes. In the context of grading essays, the concept of critical thinking could be split into sub-concepts like evidence use, logical reasoning, quality of argument and counterargument. This enables CAMA to grasp the academic assessment structure rather than just raw text generated by students.

Case-based knowledge refers to the use of historical data to resolve similar current problems with proper reasoning (IONOS, 2025). In CAMA, this encompasses past appeal results, example of human-reviewed grading, past false-positive proctoring flags, and academic integrity precedents. If a student has been flagged for noise in an online exam, CAMA can review previous cases where background noises occurred from shared residency or family activities and not due to malicious intent. This allows the system to minimize unjust punishments placed on students for factors beyond their control.

Lastly, **uncertainty-based knowledge** (Milvus, 2026) depicts the confidence levels, conflicting evidence, missing information, and risk levels of the system. For instance, during an online exam, if a student's look away from the screen, that does not necessarily constitute cheating. Uncertainty is present in CAMA, allowing it to be cautious when dealing with weak or inconclusive evidence.

2.1.2 Suitability of Knowledge Types

This four-tier approach is appropriate as education is often structure and context-dependent. The relationship in *Figure 12* supports the suitability of this approach, as the four knowledge types are not isolated but combined to produce an explainable recommendation. Rule-based and ontological knowledge provide the formal side of the system to ensure that CAMA adheres to approved rubrics, academic policies and appeal procedures. This facilitates system accountability, as its suggestions can be tracked back to well established institutional criteria instead of hidden AI presumptions (GeeksforGeeks, 2025).

The contextual side of the system is provided by case-based and uncertainty-based knowledge. They enable CAMA to make comparisons between new situations and previous cases reviewed by humans and identify weak, conflicting, or incomplete evidence. This is vital because when evidence is ambiguous, high stakes academic decisions should be made on a case-by-case basis.

Collectively, these types of knowledge help mitigate the limitations of relying on a single knowledge type. A balance must be struck between such approaches; a rule-based system can be too strict, and a purely AI-driven system can be too opaque (GeeksforGeeks, 2025). By unifying said concepts together, CAMA can generate recommendations that are explainable, thoughtful, and appropriate for human analysis.

2.2 Reasoning or Planning Approach

2.2.1 Decision-Making Under Uncertainty

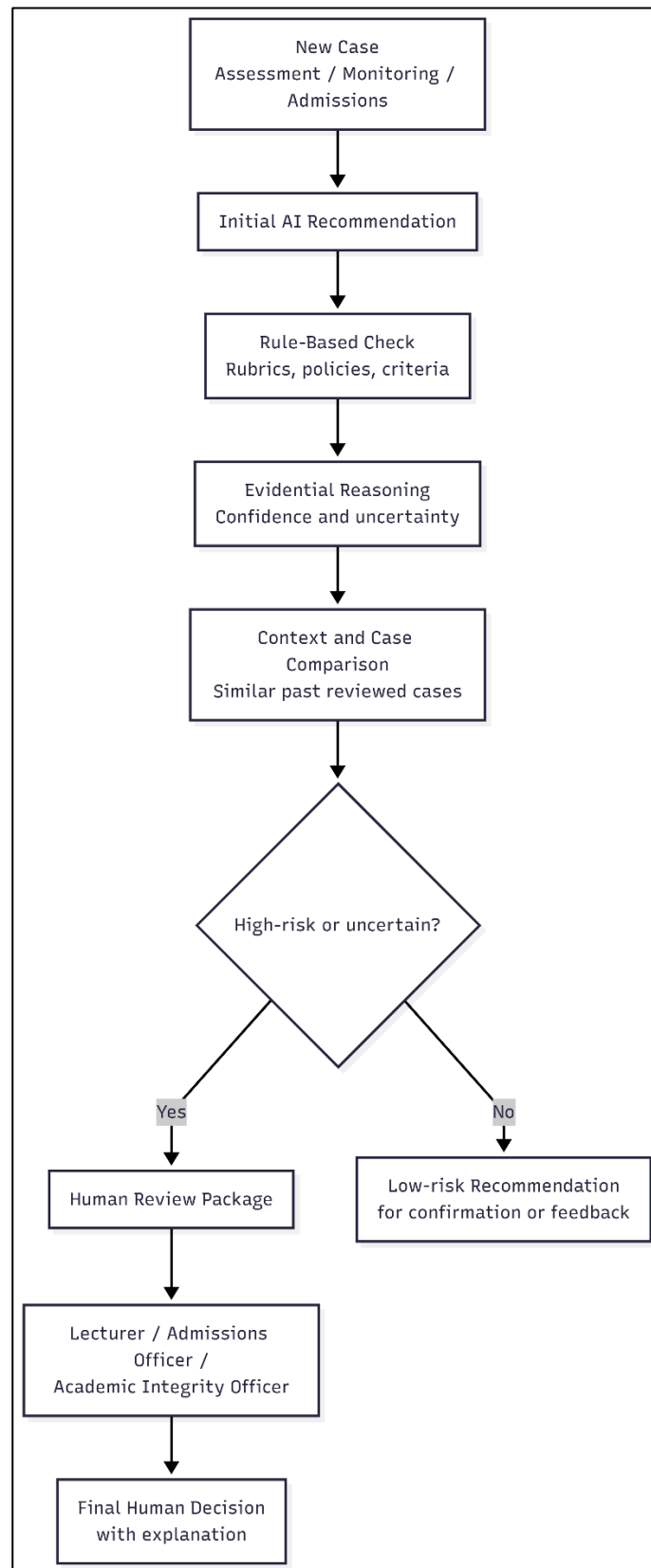


Figure 15: Human-in-the-Loop Reasoning Process in CAMA

As seen in *Figure 13*, CAMA's approach is based on a hybrid reasoning method that involves **human-in-the-loop**, **rule-based reasoning**, and **evidential reasoning**. This is essential because educational information is not necessarily accurate, as AI systems can give helpful clues, but these do not mean that a student deserves a particular mark or is guilty of wrongdoing. In the case of remote proctoring, for instance, the system may detect background noise, an unusual eye movement, low lighting, or a brief camera disconnection (Tweissi, Al Etaiwi and Al Eisawi, 2022). But these indicators could be due to co-housing, family members, internet connection, disabilities, cultural or religious matters like face coverings. If such factors are perceived as evidence of cheating, they can lead to inequitable results.

In order to deal with this, CAMA employs **evidential-reasoning** based on **Dempster-Shafer Theory**, which tackles decision-making uncertainties by combining multiple sources and assigning signify belief, doubt and lack of evidence (GeeksforGeeks, 2025). Rather than having to reach a firm conclusion, CAMA can demonstrate by presenting partial evidence of misconduct, innocence, and some degree of uncertainty. This can also be helpful when grading essays. A student may write in a creative structure, use multi-language expression, or write in a non-standard system of writing that does not conform to the pattern of the system. Instead of automatically reducing the grade, CAMA reviews the work against the rubric and previous cases and against confidence thresholds. When the recommendation is not certain, it is taken to a lecturer to be reviewed.

This mode of reasoning is based on a **caution first principle**. If the evidence is lacking, inconclusive, or counterintuitive, the system relies on human review over automated enforcement (GeeksforGeeks, 2025). In admissions, a low-confidence applicant ranking or rejection recommendation should be escalated to a human admissions officer instead of being automatically applied. This helps to minimize the chance of mis-identifying a student, grading inaccurately, and overconfident AI judgments.

2.2.2 Role of Human Oversight

Human oversight is at the core of CAMA, as the system is based on **Human-in-the-loop** principles, which means humans are not an afterthought, but rather an integral part of the decision-making process (Stryker, 2025). In addition, the escalation stage in *Figure 13* shows that human oversight is built into the reasoning process rather than added only after the AI has made a decision. This is significant since the findings in Task 1 indicated that AI can limit student agency if they are unaware or unable to challenge decisions made by these systems.

Firstly, CAMA processes an assessment by forming a recommendation. It then evaluates these recommendations

based on rules, uncertainty levels, student context, and previous cases. Then, if the case is low-risk, CAMA may give feedback or a suggested mark to the lecturer for confirmation. Otherwise, if the case is deemed to be high risk, uncertain, or potentially harmful, it is referred to an academic integrity officer.

CAMA's reviewer subsequently receives an evaluation report that includes the AI's recommendation, confidence level, uncertainty level, relevant evidence, triggered rules, and similar past cases. This will assist the reviewer in understanding why the system is uncertain, or why a case has been flagged. The reviewer will be able to accept, edit, or reject the recommendation. If the human reviewer does not agree with CAMA, the reason is stated in the case-base for future proofing purposes. The system is also contestable as students can demand for explanations and appeals, which minimizes automation bias by prompting teachers to critically analyse what the AI has generated instead of just taking it at face value.

2.3 Search or Inference Technique

2.3.1 Appropriate Techniques

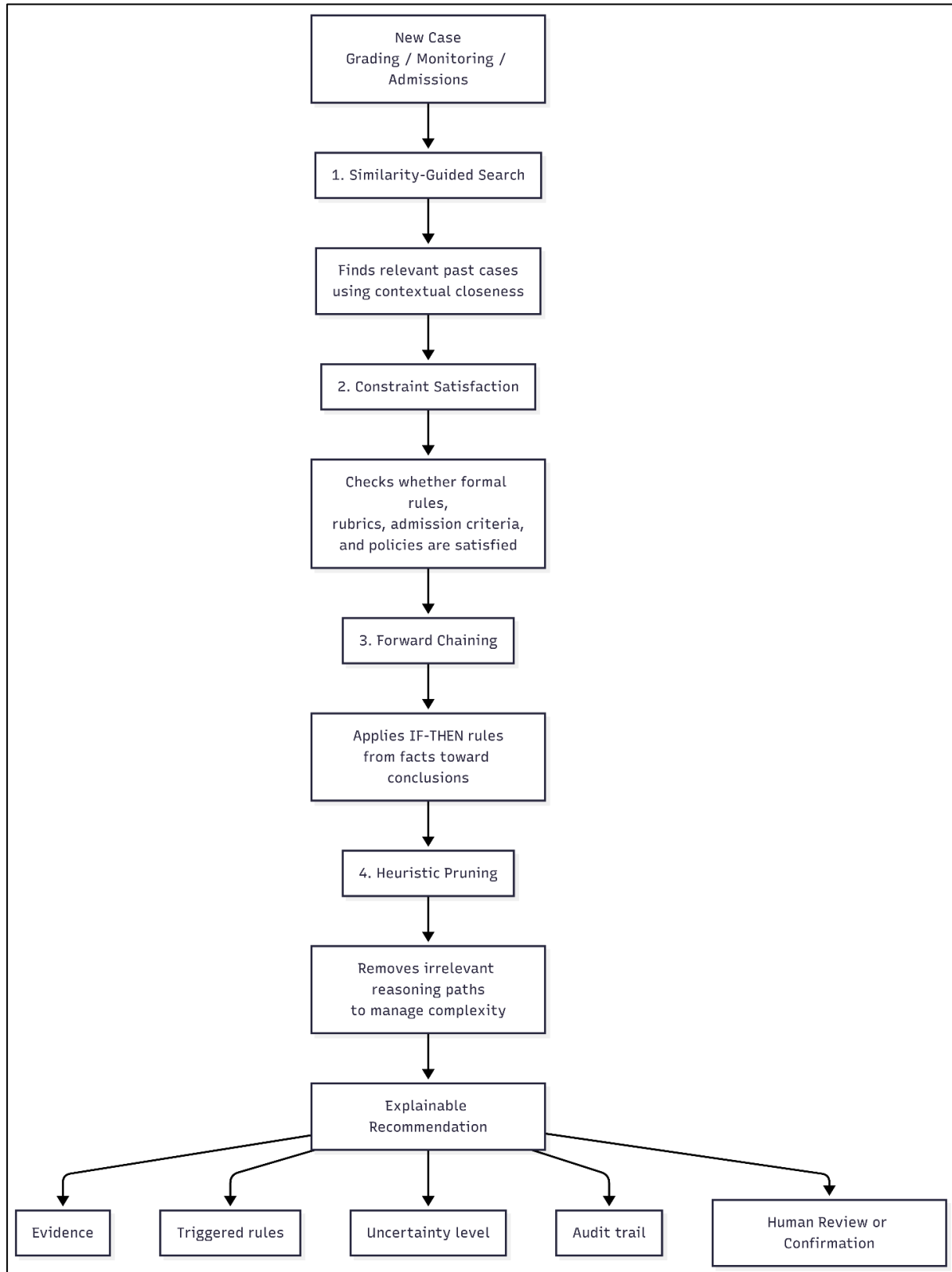


Figure 16: Search & Inference Technique Used in CAMA

Figure 14 demonstrates how CAMA implements **similarity-guided case search**, **constraint satisfaction**, **forward chaining** and **heuristic pruning**. These methods are suitable as the system should look beyond cases, follow academic rules, give explanations for conclusions, and process complex information. The first technique that will be applied is similarity-guided case search.

As new assessments or integrity cases are added into the system, CAMA will query the case-base for **similar** previous incidents. Conceptually, this can be supported using the **A* search algorithm** which uses heuristics to estimate the contextual closeness (distance) between the current and stored case (Kumar, 2024). Consideration can be given to factors such as assessment type, rubric category, integrity signal, technical issues, and previous appeal outcome. For admissions-related decisions, the same search process can retrieve similar past applicant cases based on approved criteria such as programme requirements, academic background, supporting documents, and previous human-reviewed admission outcomes. This is a good technique to use because fairness is subjective; two students might both be flagged for background noise, one due to shared household and the other from repeated suspicious communication (Tajouri and Tsoukiàs, 2024). Similarity-guided search prevents CAMA from categorizing two distinct situations as the equivalent.

The second technique is **constraint satisfaction**, which is a model that assigns values to a set of variables while obeying a specific set of rules or limitations (Tutorialspoint, 2026). In CAMA, the grading rubric and institutional policies are conditions to be met before a recommendation is accepted. For instance, a high essay grade should involve the mastery of relevance, analysis, use of sources, structure, and originality (Oxford University, 2021). In cases of academic integrity, the evidence must meet policy and human review requirements before it a misconduct recommendation can be advanced. This approach is ideal as it avoids baseless recommendations and helps minimize the risk of giving marks solely “style” rather than “substance” as all constraints are fulfilled (Andhale, 2024). Furthermore, it also ensures that harsh conclusions are not reached based on inadequate evidence.

CAMA applies **forward chaining** by using **IF-THEN rules** to draw conclusions from known facts of the students’ submission or proctoring record (Yourstory, 2026). In other words, if the similarity score is high, citations are weak, and the policy needs to be reviewed, the system initiates an academic integrity review rather than automatically issuing a penalty. Forward chaining is appropriate since academic evaluation is evidence-based and also establishes a logical audit trail (Qi, 2023). Each rule that was triggered will be documented so that the university can explain and justify its decision on why a recommendation was made, directly supporting the students’ right of appeal.

Lastly, if too many cases bases and complex rubrics generate infinite paths, **heuristic pruning** eliminates those unimportant branches fast (Sass et al., 2025). When the case is obviously not related to academic integrity, then the system will not look for academic integrity precedents.

2.3.2 Suitability in Education Context

As summarised in *Figure 14*, the appropriateness of these techniques in the education context lies in their use as a combined decision pathway, not as individual tools. First, CAMA searches for relevant patterns, compares with formal limitations, applies logic and eliminates irrelevant reasoning paths. This enables the system to present a recommendation based on admissions, monitoring, or assessment data, rather than just a raw, irrefutable assessments.

Combining these methods make CAMA more contestable than traditional black-box grading or proctoring tools. They also lessen privacy threats, as only relevant assessment or admissions evidence is required, not unlimited surveillance data. For instance, CAMA does not record every single irrelevant environmental detail, but only evidence corresponding to the approved academic rules and procedures for review and decision criteria.

For Malaysia, CAMA assists in promoting the responsible use of AI in higher education by streamlining the process of auditing automated recommendations (Malaysia.gov.my, 2026). With such a system, institutions can rely on AI to refine their operations yet still have the opportunity to review policies and decisions that impact student rights and academic trajectory.

2.4 Conclusion

CAMA demonstrates the potential of AI in the field of education without giving technology the final say over the learner. Its key innovation is that it transforms AI from a black-box decision maker to a decision support system for high-stakes academic decisions that can be reviewed.

Rather than being quick to the job, CAMA emphasizes responsible decision making. It offers the possibility for universities to leverage the power of AI for grading, monitoring, and admissions, all while ensuring that students are protected by offering explanations, contestability, and human oversight. This makes CAMA a suitable conceptual framework for mitigating ethical and social concerns and for promoting responsible adoption of AI technologies in higher education.

References

1. Asia Business Law Journal (2022). *A comparison of data protection laws*. [online] Law.asia. Available at: <https://law.asia/comparison-data-protection-laws/> [Accessed 17 May 2026].
2. Andhale, A. (2024). *Introduction to Constraint Satisfaction Problem*. [online] Medium. Available at: <https://medium.com/@andy08/introduction-to-constraint-satisfaction-problem-93c476997cd2> [Accessed 16 May 2026].
3. Bernik, A., Radošević, D. and Čep, A. (2025). A Comparative Study of Large Language Models in Programming Education: Accuracy, Efficiency, and Feedback in Student Assignment Grading. *Applied Sciences*, [online] 15(18), p.10055. <https://doi.org/10.3390/app151810055>.
4. Bhatt, T. (2023). *AI in Education: Uses, Benefits, Examples & More*. [online] Intelivita. Available at: <https://www.intelivita.com/blog/ai-in-education/> [Accessed 21 Apr. 2026].
5. Bird, K.A., Castleman, B.L., Mabel, Z. and Song, Y. (2021). Bringing Transparency to Predictive Analytics: A Systematic Comparison of Predictive Modeling Methods in Higher Education. *AERA Open*, [online] 7. <https://doi.org/10.1177/23328584211037630>. [Accessed 16 May 2026].
6. Chaudhary, A. (2025). *Explainable AI: Demystifying the Black Box*. [online] Medium. Available at: <https://medium.com/@arjun.chaudhary.gnd/explainable-ai-demystifying-the-black-box-4b845da14b32> [Accessed 16 May 2026].
7. Cordella, A. and Gualdi, F. (2026). Historical Homologation in AI Algorithmic Computation: When the Past Decides Your Future. *Proceedings of the Annual Hawaii International Conference on System Sciences*. [online] ResearchGate. Available at: https://www.researchgate.net/publication/399553410_Historical_homologation_in_AI_algorithmic_computation_when_the_past_decides_your_future [Accessed 16 May 2026].

8. Crowell. (2025). *Ivy League Lawsuit Centers on Alleged Impermissible Use of AI in Academia*. [online] Available at: <https://www.crowell.com/en/insights/client-alerts/ivy-league-lawsuit-centers-on-alleged-impermissible-use-of-ai-in-academia> [Accessed 16 May 2026].
9. Diplo. (2023). *Universities stop using AI detection tool such as Turnitin – Diplo*. [online] Available at: <https://www.diplomacy.edu/updates/universities-stop-using-ai-detection-tool-such-as-turnitin/> [Accessed 16 May 2026].
10. Doss, C.J., Bozick, R., Schwartz, H.L., Chu, L., Rainey, L.R., Woo, A., Reich, J. and Dukes, J. (2025). *AI Use in Schools Is Quickly Increasing but Guidance Lags Behind: Findings from the RAND Survey Panels*. [online] Rand.org. Available at: https://www.rand.org/pubs/research_reports/RRA4180-1.html [Accessed 28 Apr. 2026].
11. EduTech (2026). *AI in University Admissions: Impact, Benefits & Future Trends*. [online] EduTech. Available at: <https://edutech.global/ai-in-university-admissions/> [Accessed 16 May 2026].
12. ETS (2026). *How the e-rater Scoring Engine Works*. [online] Ets. Available at: <https://www.ets.org/erater/how.html> [Accessed 16 May 2026].
13. European Commission (2026). *AI Act: Shaping Europe’s Digital Future*. [online] Shaping Europe’s digital future. Available at: <https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai> [Accessed 16 May 2026].
14. ExamSoft. (2020). *ExamSoft*. [online] Available at: <https://examsoft.com/> [Accessed 17 May 2026].
15. GeeksforGeeks (2025). *ML | Dempster Shafer Theory*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/machine-learning/ml-dempster-shafer-theory/> [Accessed 16 May 2026].
16. GeeksforGeeks (2025). *Rule-Based System in AI*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/artificial-intelligence/rule-based-system-in-ai/> [Accessed 16 May 2026].
17. Gemini. (2025). *Gemini for Students — your AI study buddy from Google*. [online] Available at:

<https://gemini.google/students/> [Accessed 16 May 2026].

18. Gullo, K. (2023). *After Students Challenged Proctoring Software, French Court Slaps TestWe App With a Suspension*. [online] Electronic Frontier Foundation. Available at: <https://www.eff.org/deeplinks/2023/03/after-students-challenged-proctoring-software-french-court-slaps-testwe-app> [Accessed 16 May 2026].
19. Han, T. and Zheng, Y. (2026). AI and human scoring for postgraduate writing: Evaluating score reliability, variability, and rater behaviours. *Studies in Educational Evaluation*, [online] 88, p.101572. <https://doi.org/10.1016/j.stueduc.2026.101572>. [Accessed 16 May 2026].
20. Hansel, C.A., Ottenbreit-Leftwich, A., Quick, J.D., Greene, A.H. and Ricci, M. (2024). Gradescope in Large Lecture Classrooms: A Case Study at Indiana University. *Journal of Teaching and Learning with Technology*, [online] 13(1). <https://doi.org/10.14434/jotlt.v13i1.38519>. [Accessed 16 May 2026].
21. INFOLKS (2026). LinkedIn. [online] Linkedin.com. Available at: <https://www.linkedin.com/pulse/llm-as-a-judge-vs-human-evaluation-infolks-fnscc/> [Accessed 6 May 2026].
22. IONOS (2025). *What is case-based reasoning?* [online] ionos Digital Guide. Available at: <https://www.ionos.com/digitalguide/websites/web-development/case-based-reasoning/> [Accessed 16 May 2026].
23. Islam, N. (2025). *Scaling Evaluation with LLM Judges: Our Approach and Findings*. [online] Medium. Available at: <https://medium.com/@nomannayeem/scaling-evaluation-with-llm-judges-our-approach-and-findings-0a046e8344c4> [Accessed 28 Apr. 2026].
24. Jhunjhunwala, M.S. (2026). LinkedIn. [online] Linkedin.com. Available at: <https://www.linkedin.com/pulse/start-ups-over-reliance-technology-manesh-saroj-jhunjhunwala/> [Accessed 17 May 2026].
25. Jiang, T., Er, S., Heron, M.J., Zhu, K.J. and Yang, R. (2025). Artificial Intelligence Use in Academic Applicant Screening: A Systematic Review. *Plastic and Reconstructive Surgery - Global Open*, [online] 13(10), p.e7177. <https://doi.org/10.1097/gox.0000000000007177>. [Accessed 17 May 2026].

26. Karthiga, J. (2026). *LinkedIn*. [online] LinkedIn.com. Available at: <https://www.linkedin.com/pulse/black-box-breakthrough-demystifying-ai-trust-jyothi-karthiga-dpqac/> [Accessed 16 May 2026].
27. Kelly, A. (2021). *The great algorithm fiasco*. [online] Bera. Available at: <https://www.bera.ac.uk/blog/the-great-algorithm-fiasco> [Accessed 16 May 2026].
28. Kenenhan, S. (2025). *QWK scores: Automated grading system efficacy | Learnosity*. [online] Learnosity.com. Available at: <https://learnosity.com/ai-assisted-scoring-feedback/automated-grading-system-qwk-scores/> [Accessed 28 Apr. 2026].
29. Kenton, W. (2025). *Understanding Black Box Models: Definition, Finance Use, and Examples*. [online] Investopedia. Available at: <https://www.investopedia.com/terms/b/blackbox.asp> [Accessed 17 May 2026].
30. Kosinski, M. (2024). *Black box AI*. [online] Ibm.com. Available at: <https://www.ibm.com/think/topics/black-box-ai> [Accessed 16 May 2026].
31. Kumar, R. (2024). *The A* Algorithm: A Complete Guide*. [online] Datacamp.com. Available at: <https://www.datacamp.com/tutorial/a-star-algorithm> [Accessed 16 May 2026].
32. Kwapisz, M.B., Ackerman, Y., Nguyen, J. and Rajivan, P. (2025). *Surveillance and Disability in Online Proctored Exams: Student Perspectives and Design Implications*. Arxiv.org. [online] <https://doi.org/10.1145/nnnnnnn>. [Accessed 16 May 2026].
33. Leadersinlaw (2026). *Leaders in Law*. [online] Leaders in Law. Available at: <https://www.leaders-in-law.com/how-data-protection-laws-shape-university-admissions-processes/> [Accessed 16 May 2026].
34. Liaison. (2026). *Othot | Predictive Analytics Tool for Higher Education*. [online] Available at: <https://www.liaisonedu.com/higher-education-ai/othot/> [Accessed 16 May 2026].
35. Malaysia.gov.my. (2026). *National Guidelines on Artificial Intelligence Governance and Ethics (AIGE)*. [online] Available at: <https://www.malaysia.gov.my/en/my-initiative/whole-government-digital-services/tadbir-urus/national-guidelines-on-artificial-intelligence-governance-and-ethics-aige> [Accessed 16 May 2026].

36. MIDA (2025). *AI in Education Driving Malaysia's Future-Ready Workforce*. [online] MIDA | Malaysian Investment Development Authority. Available at: <https://www.mida.gov.my/ai-in-education-driving-malaysias-future-ready-workforce/> [Accessed 16 May 2026].
37. Miller, C.L. and Manderfeld, M. (2021). *Online Proctoring Considerations*. [online] Maverick Learning and Educational Applied Research Nexus. Available at: <https://mlpp.pressbooks.pub/mavlearn/chapter/online-proctoring-considerations/> [Accessed 16 May 2026].
38. Milvus (2026). *What is uncertainty reasoning in AI?* [online] Milvus.io. Available at: <https://milvus.io/ai-quick-reference/what-is-uncertainty-reasoning-in-ai> [Accessed 16 May 2026].
39. Mit.edu. (2026). *Right to Contest in AI Blindspot - A Discovery Process for preventing, detecting, and mitigating bias in AI systems*. [online] Available at: https://aiblindspot.media.mit.edu/right_to_contest.html [Accessed 17 May 2026].
40. Mita, S. (2023) *AI Proctoring: Academic Integrity vs Student Rights*. Available at: https://repository.uclawsf.edu/cgi/viewcontent.cgi?article=4044&context=hastings_law_journal [Accessed 16 May 2026].
41. Mullainathan, S. (2019). *Biased Algorithms Are Easier to Fix Than Biased People*. [online] Nytimes.com. Available at: <https://www.nytimes.com/2019/12/06/business/algorithm-bias-fix.html> [Accessed 5 May 2026].
42. NAI0 (2025). *Building an AI Nation Together, Now*. [online] Ai.gov.my. Available at: <https://ai.gov.my/> [Accessed 16 May 2026].
43. OECD. (2026). *AI principles*. [online] Available at: <https://www.oecd.org/en/topics/sub-issues/ai-principles.html> [Accessed 16 May 2026].
44. Oxford University. (2021). *Essay and dissertation writing skills*. [online] Available at: <https://www.ox.ac.uk/students/academic/guidance/skills/essay> [Accessed 16 May 2026].
45. Parva Consulting (2025). *The European Union Artificial Intelligence Act: Bridging ethics, innovation*

- and governance. [online] Parva Consulting. Available at: <https://parvaconsulting.com/articles/eu-ai-act-ethics-innovation-governance-explained/> [Accessed 17 May 2026].
46. PDPD (2026) *AUTOMATED DECISION-MAKING AND PROFILING (ADMP)* [online] pdp.gov.my. Available at: <https://www.pdp.gov.my/ppdpv1/wp-content/uploads/2026/04/Automated-Decision-Making-And-Profiling-Guideline-ADMP.pdf> [Accessed 16 May 2026].
47. Pogożelski, E. (2023). *Inequalities in College Admissions - Sharing Sociology*. [online] Sharing Sociology. Available at: <https://soc.hamiltonlits.org/sharingsociology/fall-2022/inequalities-in-college-admissions/> [Accessed 16 May 2026].
48. Popov, P. (2026). *The Power of Ontologies and Knowledge Graphs: Practical Examples from the Financial Industry*. [online] Graphwise. Available at: <https://graphwise.ai/blog/the-power-of-ontologies-and-knowledge-graphs-practical-examples-from-the-financial-industry/> [Accessed 16 May 2026].
49. Proctored Exams: Student Perspectives and Design Implications. *Arxiv.org*. [online] <https://arxiv.org/pdf/2511.10826> [Accessed 16 May 2026].
50. Proctorio. (2026). *Securing the integrity of online assessments | Proctorio*. [online] Available at: <https://proctorio.com/> [Accessed 28 Apr. 2026].
51. ProctorU. (2025). *The ProctorU Proctoring Platform - Advanced Exam Technology Backed by Human Validation*. [online] Available at: <https://www.proctoru.com/> [Accessed 17 May 2026].
52. Qi, T. (2023). Exploration of the Concept of “Evidence-Based Evaluation” in Classroom Teaching Evaluation —Taking “Literature Reading and Writing” Task Group Practice as an Example. *Open Journal of Social Sciences*, [online] 11(10), pp.640–648. <https://doi.org/10.4236/jss.2023.1110037>. [Accessed 16 May 2026].
53. Rhoads, T. (2024). *Top 10 Surprises of Accountability - Greater Reading Chamber Alliance*. [online] Greater Reading Chamber Alliance. Available at: <https://greaterreading.org/top-10-surprises-of-accountability/> [Accessed 16 May 2026].
54. Rubman, J. (2024). *AI-Assisted Grading: A Magic Wand or a Pandora’s Box? - MIT Sloan Teaching*

- & Learning Technologies. [online] MIT Sloan Teaching & Learning Technologies. Available at: <https://mitsloanedtech.mit.edu/2024/05/09/ai-assisted-grading-a-magic-wand-or-a-pandoras-box/> [Accessed 16 May 2026].
55. Sass, S.-H., Gönner, L., Schwöbel, S., Frölich, S., Glöckner, F., Kiebel, S.J., Li, S.-C. and Smolka, M.N. (2025). Heuristic pruning of decision trees at low probabilities and probability discounting in sequential planning in young and older adults. *Scientific Reports*, [online] 15(1). <https://doi.org/10.1038/s41598-025-00905-7>.
 56. Securiti. (2024). *An Overview of Malaysia's National Guidelines on AI Governance and Ethics - Securiti*. [online] Available at: <https://securiti.ai/malaysia-national-guidelines-on-ai-governance-and-ethics/> [Accessed 16 May 2026].
 57. Selvam, M. and González Vallejo, R. (2025). Human-in-the-Loop Models for Ethical AI Grading: Combining AI Speed with Human Ethical Oversight. *EthAlca*, [online] 4, p.413. <https://doi.org/10.56294/ai2025413>. [Accessed 16 May 2026].
 58. Shaak, E. (2021). *Online Test-Taking Software ProctorU Violates Ill. Privacy Law, Class Action Alleges*. [online] Classaction.org. Available at: <https://www.classaction.org/news/online-test-taking-software-proctoru-violates-ill-privacy-law-class-action-alleges> [Accessed 16 May 2026].
 59. Shah, E. (2023). *Scaler*. [online] Scaler Topics. Available at: <https://www.scaler.com/topics/artificial-intelligence-tutorial/rule-based-system-in-ai/> [Accessed 16 May 2026].
 60. Shutterstock. (2022). *Student Cheating Academic Exam Girl Cheater Stock Vector (Royalty Free) 2202066515 | Shutterstock*. [online] Available at: <https://www.shutterstock.com/image-vector/student-cheating-academic-exam-girl-cheater-2202066515?trackingId=842b0ab4-c14f-41cd-af30-c8842a7b24c5&listId=searchResults> [Accessed 16 May 2026].
 61. Stempel, J. (2024). *SAT administrator College Board settles New York claims it sold student data*. [online] Reuters. Available at: <https://www.reuters.com/legal/sat-administrator-college-board-settles-new-york-claims-it-sold-student-data-2024-02-13/> [Accessed 16 May 2026].
 62. Stryker, C. (2025). *Human In The Loop*. [online] Ibm.com. Available at: <https://www.ibm.com/think/topics/human-in-the-loop> [Accessed 16 May 2026].

63. Sun, V. (2026). *7 Best AI Tools for Student Enrollment in Higher Education | Wonderchat: AI Chatbots for your website*. [online] Wonderchat.io. Available at: <https://wonderchat.io/blog/ai-enrollment-tools-students> [Accessed 28 Apr. 2026].
64. Tajouri, S. and Tsoukiàs, A. (2024). *Subjective fairness in algorithmic decision-support*. [online] arXiv.org. Available at: <https://arxiv.org/abs/2407.01617> [Accessed 16 May 2026].
65. Talbot, D. (2016). *The Unacceptable Persistence of the Digital Divide*. [online] MIT Technology Review. Available at: <https://www.technologyreview.com/2016/12/16/155240/the-unacceptable-persistence-of-the-digital-divide/> [Accessed 7 May 2026].
66. Tan, Le Y., Hu, S., Yeo, D.J. and Cheong, K.H. (2025). A Comprehensive Review on Automated Grading Systems in STEM Using AI Techniques. *Mathematics*, [online] 13(17), p.2828. <https://doi.org/10.3390/math13172828>. [Accessed 16 May 2026].
67. Turnitin (2025). *New and emerging trends in academic misconduct | Turnitin*. [online] Turnitin. Available at: <https://www.turnitin.com/blog/what-are-the-new-and-emerging-trends-in-academic-misconduct> [Accessed 16 May 2026].
68. Turnitin (2025). *Turnitin Feedback Studio | Student writing excellence*. [online] Turnitin.com. Available at: <https://www.turnitin.com/products/feedback-studio/> [Accessed 16 May 2026].
69. Turnitin (2026). *AI-native academic integrity trends and impact | Turnitin*. [online] Available at: <https://www.turnitin.com/blog/are-universities-ready-for-ai-native-academic-integrity> [Accessed 6 May 2026].
70. Tweissi, A., Al Etaiwi, W. and Al Eisawi, D. (2022). The Accuracy of AI-Based Automatic Proctoring in Online Exams. *Electronic Journal of e-Learning*, [online] 20(4), pp.419–435. <https://doi.org/10.34190/ejel.20.4.2600>.
71. Unesco (2026). *Artificial intelligence in education*. [online] Unesco.org. Available at: <https://www.unesco.org/en/digital-education/artificial-intelligence> [Accessed 16 May 2026].
72. University Canada West (2025). *Advantages and disadvantages of AI in education*. [online]

- Ucanwest.ca. Available at: <https://www.ucanwest.ca/blog/education-careers-tips/advantages-and-disadvantages-of-ai-in-education> [Accessed 28 Apr. 2026].
73. Vasquez, C., Gandara, D., Sánchez, Zorayda. and Nuñez, Sergio. (2025) Selling Student Success: A Critical Analysis of Predictive Analytics Vendors in Higher Education. *EdWorkingPaper No. 25-1349*, [online] <https://edworkingpapers.com/sites/default/files/ai25-1349.pdf> [Accessed 16 May 2026].
 74. Vecteezy (2026). *Authoritarian regime flat vector illustration. Political system metaphor. Form of government. Manipulating and controlling institutions, economics. Centralized power cartoon characters.* [online] Vecteezy. Available at: <https://www.vecteezy.com/vector-art/4244955-authoritarian-regime-flat-vector-illustration-political-system-metaphor-form-of-government-manipulating-and-controlling-institutions-economics-centralized-power-cartoon-characters> [Accessed 17 May 2026].
 75. Watermark Insights (2024). *How predictive analytics helps improve student enrollment.* [online] Watermark Insights. Available at: <https://www.watermarkinsights.com/resources/blog/how-predictive-analytics-helps-improve-student-enrollment/> [Accessed 16 May 2026].
 76. Wharton (2023). *How Data Privacy Concerns Impact Firm Performance.* [online] Knowledge at Wharton. Available at: <https://knowledge.wharton.upenn.edu/article/how-data-privacy-concerns-impact-firm-performance/> [Accessed 17 May 2026].
 77. Yamkovenko, S. (2024). *8 Reasons Every Parent Should Consider AI Tutor Khanmigo for Their Child's Learning.* [online] Khan Academy Blog. Available at: <https://blog.khanacademy.org/8-reasons-every-parent-should-consider-khanmigo-for-their-childs-learning-kp/> [Accessed 16 May 2026].
 78. Yoder-Himes, D.R., Asif, A., Kinney, K., Brandt, T.J., Cecil, R.E., Himes, P.R., Cashon, C., Hopp, R.M.P. and Ross, E. (2022). Racial, skin tone, and sex disparities in automated proctoring software. *Frontiers in Education*, [online] 7. <https://doi.org/10.3389/feduc.2022.881449>.
 79. Yourstory (2026). *Forward Chaining.* [online] Available at: <https://yourstory.com/glossary/forward-chaining> [Accessed 16 May 2026].
 80. Young, L. (2025). *New and emerging trends in academic misconduct | Turnitin.* [online] Turnitin.

Available at: <https://www.turnitin.com/blog/what-are-the-new-and-emerging-trends-in-academic-misconduct> [Accessed 16 May 2026].

81. Yulianti, S. (2025). *The Hidden Bias in AI: How Artificial Intelligence Reflects and Reinforces Social Inequalities*. [online] ResearchGate. Available at: https://www.researchgate.net/publication/389895420_The_Hidden_Bias_in_AI_How_Artificial_Intelligence_Reflects_and_Reinforces_Social_Inequalities [Accessed 16 May 2026].
82. Zahn, B. (2025). *Yale Student Suing Over Accusation of Improper AI Use*. [online] GovTech. Available at: <https://www.govtech.com/education/higher-ed/yale-student-suing-over-accusation-of-improper-ai-use>. [Accessed 16 May 2026].

Appendix

Turnitin Draft (Task 1 & Task 2).docx			
ORIGINALITY REPORT			
2%	2%	1%	0%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS
PRIMARY SOURCES			
1	philarchive.org Internet Source	<1 %	
2	Akanksha Srivastava, A.V. Senthil Kumar, Disha Maheshwari, Samrat Ray. "Artificial Intelligence and Human Existence - Insights on Education, Technology, and Society", Routledge, 2025 Publication	<1 %	
3	Submitted to International University of Malaya-Wales Student Paper	<1 %	
4	www.eff.org Internet Source	<1 %	
5	link.springer.com Internet Source	<1 %	
6	www.klgates.com Internet Source	<1 %	
7	akasa.com.ng Internet Source	<1 %	